Grokking Under a Fixed Budget

One Long Run or Several Restarts?

Probability and Statistics | BAMAT207
Review 1 | 15 October 2026

Name: [Your name]
Registration number: [Your registration number]

Our problem and objectives

How much training does a model need to answer new questions?

Model = a learning program; here, a small neural network.

Measure how much the waiting time changes between attempts.

Compare training settings and the chance of success within a limit.

Budget = total learning steps; restart = begin again from a fresh start.

What is grokking?

The model gets practice questions right first.
After more training, it improves on new questions.

That delayed improvement is called grokking.

Memorization = doing well on the practice examples.

Generalization = doing well on questions excluded from practice.

Accuracy = the percentage of answers that are correct.

Think of learning a maths rule after practising examples. Background: Power et al. [1]

How we run the experiment

Task: add two numbers, then keep the remainder after dividing by 31.
(30 + 4) mod 31 = 3 because 34 = 31 + 3.

961 number pairs → 576 practice questions + 385 new questions.

Random seed = a number that chooses a repeatable starting point.

Training update = one learning step that adjusts the model.

Python runs and records the experiment; R analyses the results.

Weight decay gently shrinks the model's internal numbers; compare 0.3, 1 and 3.
24 main attempts; 9 setup attempts are excluded from these comparisons.

How we measure learning time

T = learning steps needed to pass our success rule

Get ≥ 99% of practice answers and ≥ 95% of new answers correct at three checks in a row. Check every 25 updates.

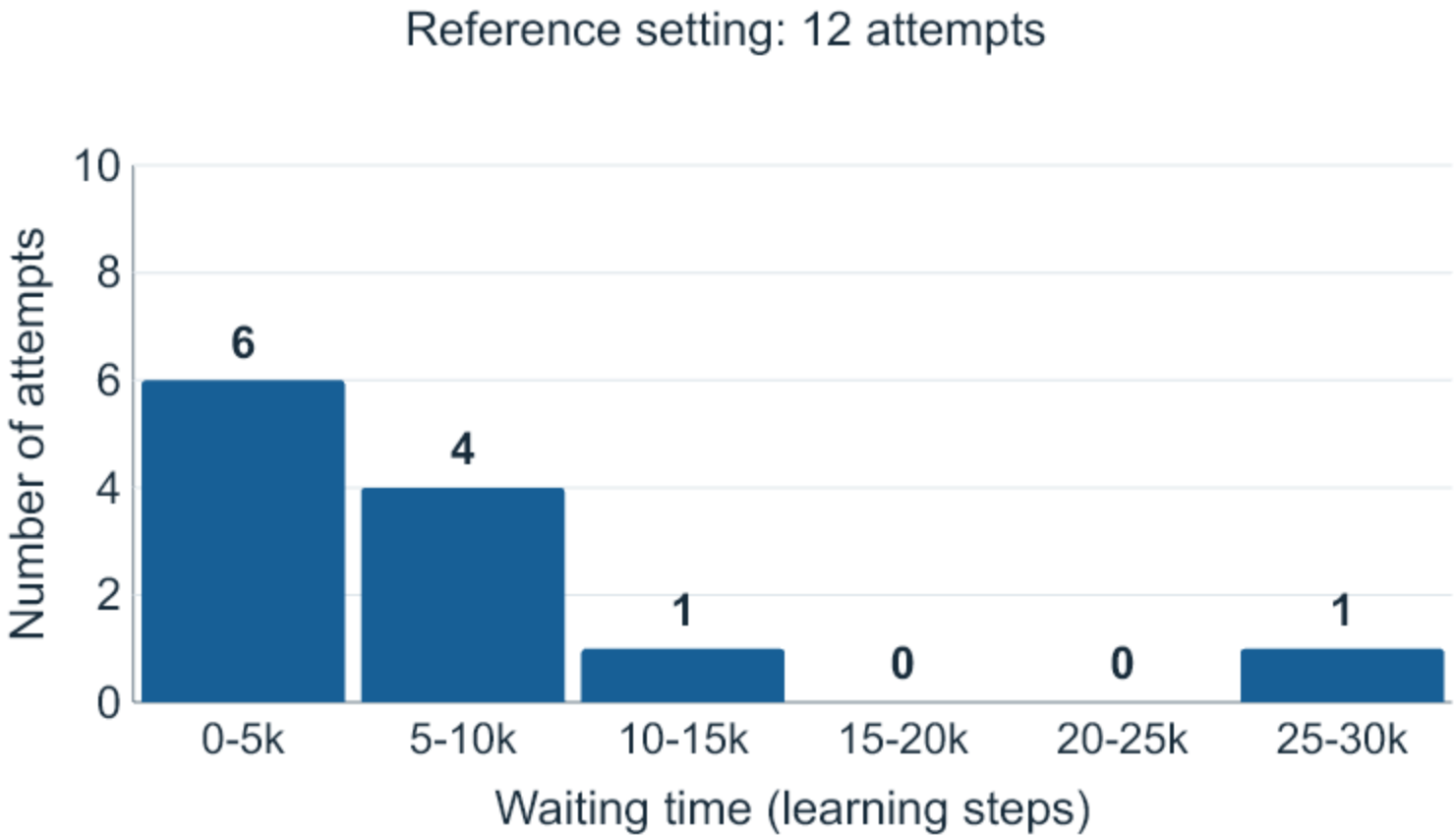
Pass at 2,000, 2,025 and 2,050 → record T = 2,050.

T changes between attempts: it is a random variable.

Limit C = 30,000. Capped time U = min(T, C).
If unfinished, eventual T is unknown; U records the limit.

Three successful checks do not guarantee that accuracy stays high forever.

How much do learning times vary?

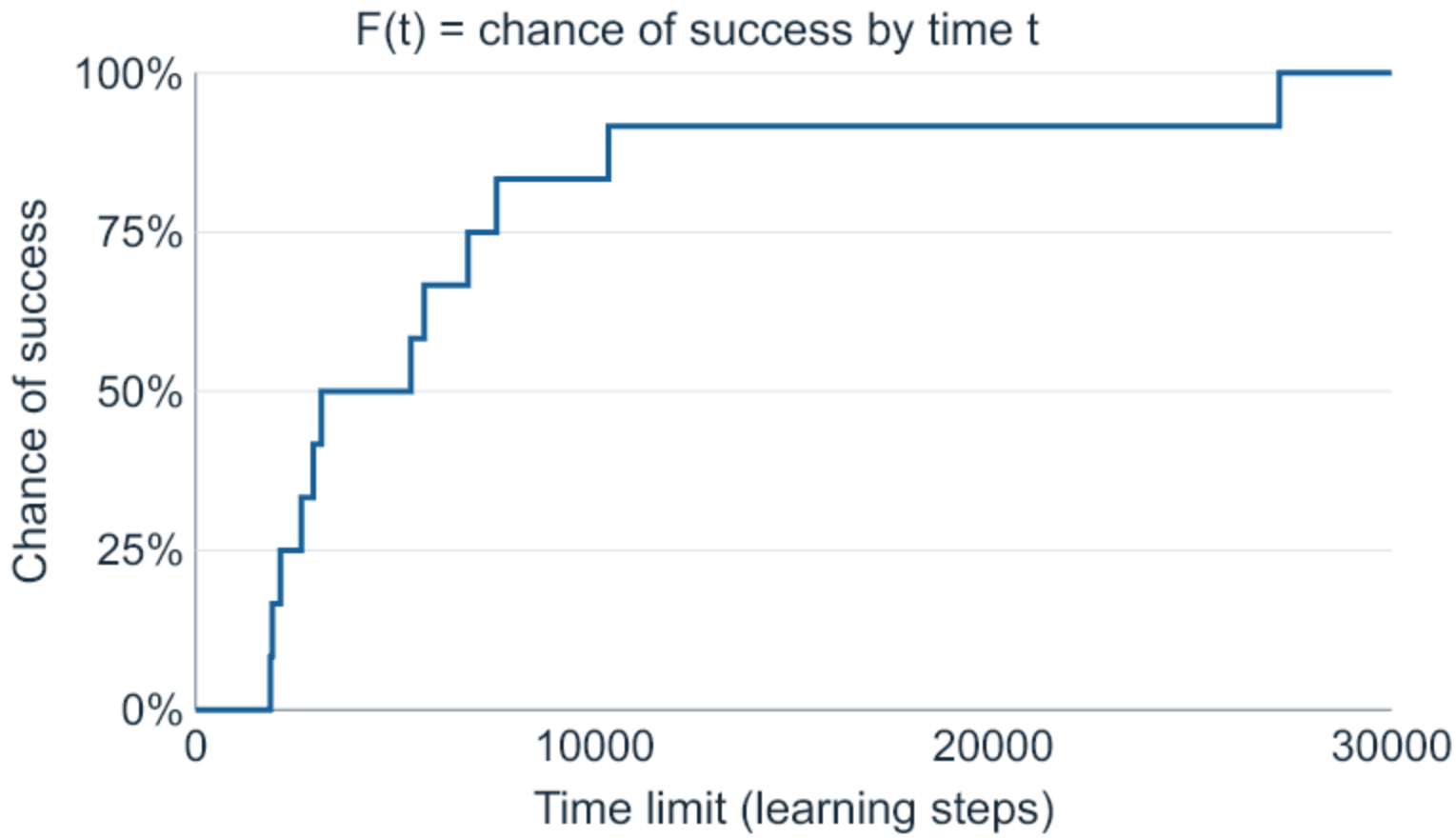


Mean (average): 6,475
Median (middle): 4,275
SD (spread): 7,041

Range:
1,875–27,175

SD = standard deviation: spread around the average. One slow run raises the mean.

What is the chance of learning in time?



By 2,400: 3/12 = 25%
By 30,000: 12/12 succeed

95% confidence interval:
73.5%–100%
Shows uncertainty from
a small sample.

Bernoulli = one yes/no result.
Binomial = number of successes.

All 12 worked by the limit. Another attempt could still fail.

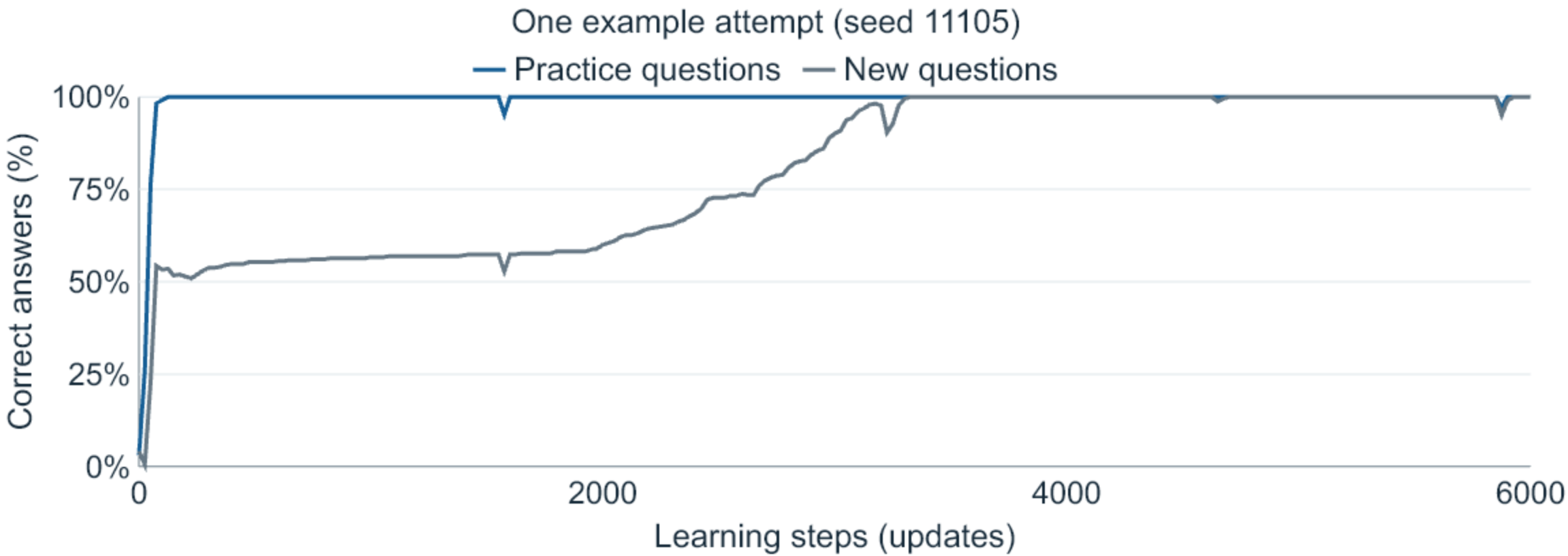
How we compare the results

Method	Plain meaning	Question answered
Welch’s t-test	Compare two group averages.	Could the gap be due to chance?
ANOVA	Compare all three averages.	Could all groups have the same true average?
Correlation	Measure how two things vary together.	Does larger decay go with less waiting?
Regression	Fit a line through the data.	How does waiting time relate to decay?

ANOVA p = 0.0077: a surprising gap if group means were equal.

Check the test’s conditions: our errors do not follow the usual bell-shaped pattern.
Treat these small-sample comparisons as early evidence.

Grokking in one training attempt

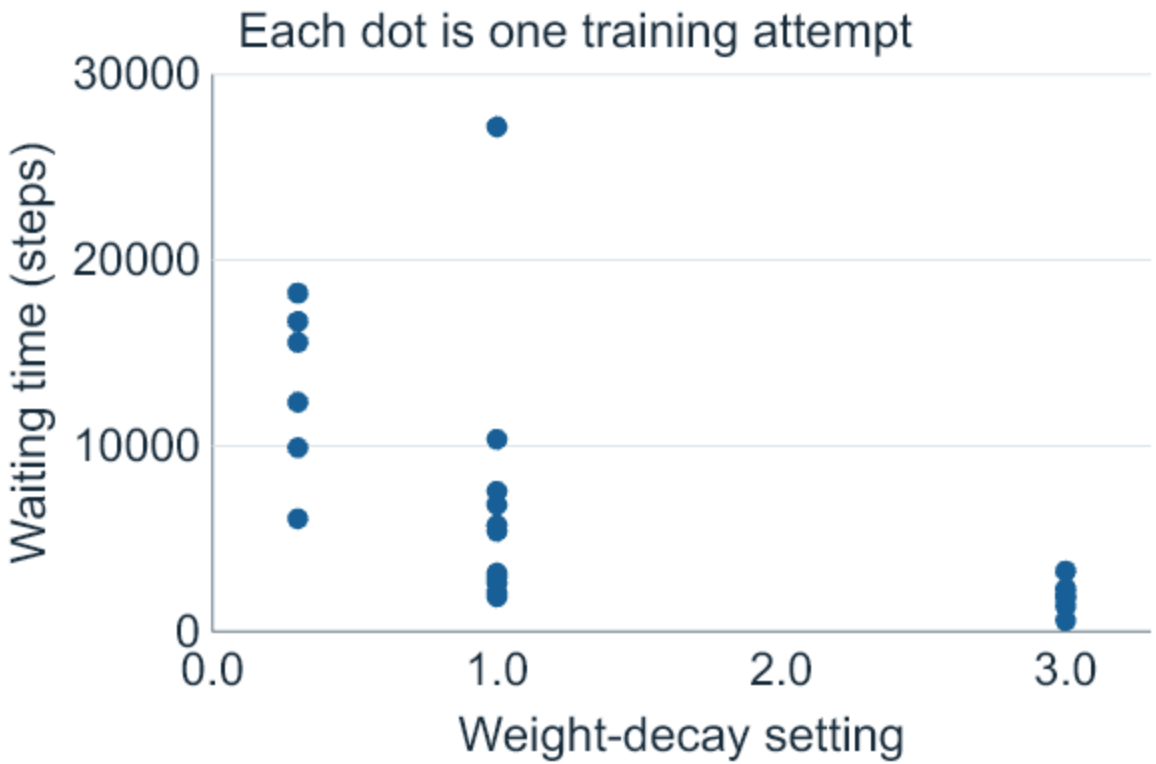


Practice success: 150 | New-question success T: 3150 steps

First 6,000 steps shown. Later accuracy drops occurred in 18 of the 24 attempts.

What happened when we changed weight decay?

Weight decay = a control that gently shrinks the model's adjustable numbers.



Setting	Runs	Average	SD
0.3	6	13,138	4,593
1	12	6,475	7,041
3	6	1,888	890

All 24 passed by 30,000 steps. Higher decay went with shorter waiting times in this sample.

Small groups; one practice/test split. More repeats are needed for broader conclusions.

One long run or several fresh starts?

$P(\text{at least one success}) = 1 - [1 - F(B/k)]^k$

B = total steps; k = attempts; F(t) = chance of success by step t.

Fresh starts (k)	Steps per attempt	Predicted chance
1 attempt	30,000	100.0%
2 attempts	15,000	99.3%
4 attempts	7,500	99.6%

Total B = 30,000. Chance at least one works = 1 − chance all fail.

Predictions from 12 reference runs; unvalidated in the Review 1 sample.

Progress and next questions

Review 1 milestone: approximately 65% of implementation

Training code, saved progress, repeat attempts, statistics and first report.

Review 2 will cover

More attempts; mathematical descriptions of waiting-time patterns; comparisons using more than one training control.

Further research

Try smaller budgets and different practice/test questions.
Check if success stays high (sustained generalization).

Where the ideas come from

[1] Power et al. (2022)
Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets

https://arxiv.org/abs/2201.02177

[2] Truong et al. (2026), preprint
First-Passage Prediction of Grokking Delay: A Calibrated Law under AdamW with Causal Validation

https://arxiv.org/abs/2605.18845

[3] Streeter, Golovin and Smith (2007)
Restart Schedules for Ensembles of Problem Instances

https://www.cs.cmu.edu/~dgolovin/papers/aaai07b.pdf

[4] Supplied BAMAT207 syllabus | [5] Project's frozen Review 1 data

Our proposed addition: measure waiting times and compare training budgets.
These earlier ideas are credited; a worldwide-first claim is not established.

Appendix A: restart probability, step by step

Illustration with assumed chances; not a measured result

Two attempts × 6,000 steps = 12,000 total steps.
Assume each has 60% success and 40% failure.

Both fail: 0.40 × 0.40 = 0.16 (16%)

At least one works: 1 − 0.16 = 0.84 (84%)

Independent = one outcome does not change the other's chance.
We still need the one-long-run chance to choose a winner.

Appendix B: useful terms and questions

Seed = repeatable starting point. Update = one learning step.

Generalization = doing well on questions excluded from practice.

Standard deviation = spread. Confidence interval = uncertainty range.

Why three checks? Reduce reliance on one lucky evaluation.

Unfinished attempt? We do not know its eventual learning time.

Limits: few attempts, one question split, test assumptions and later drops.